

Customer Churn & Retention Intelligence Dashboard

Business Intelligence Project Report

Generated: September 24, 2026

Dataset: Kaggle — Customer Churn Dataset (Training + Testing)

Source: <https://www.kaggle.com/datasets/muhammadshahidazeem/customer-churn-dataset>

Technology: Python · Streamlit · Plotly · scikit-learn · ReportLab

Purpose: Academic Internship Submission

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1. Abstract

This report presents the findings of a Business Intelligence analysis on customer churn behaviour for a subscription-based service. Using a dataset of approximately 440,832 cleaned customer records sourced from Kaggle, this project identifies the key drivers of churn, segments customers by value and risk, builds and evaluates predictive churn models, and translates findings into concrete, data-supported business recommendations. The project is delivered as an interactive Streamlit dashboard alongside this formal report. All statistics cited in this document are computed directly from the actual dataset and verified against the live application.

2. Introduction

Customer churn — the loss of subscribers or service users — is one of the most significant challenges for subscription-based businesses. Customer retention is an important business objective because reducing churn can help protect recurring revenue and customer relationships, which are often more cost-effective to maintain than to rebuild through new customer acquisition.

This project applies exploratory data analysis, customer segmentation, and predictive modelling to a real-world customer churn dataset to produce an actionable Business Intelligence solution. The findings are presented both as an interactive dashboard and as this formal analytical report.

3. Business Problem

The business problem addressed by this project is: *Which customers are at risk of churning, what behavioural patterns are associated with churn, and what data-supported actions can the business consider to reduce customer attrition?*

- What is the current churn rate, and how does it differ across customer segments?
- Which behavioural and contractual characteristics are most associated with churn?
- Which customers represent the highest concentration of churn risk?
- What specific, evidence-based actions could reduce churn?

4. Project Objectives

- Calculate accurate KPIs from the cleaned and validated dataset
- Identify churn patterns across subscription types, contract lengths, and behavioural metrics
- Build data-driven customer segments based on spend and risk indicators
- Train and evaluate predictive churn models (Logistic Regression and Random Forest)
- Translate all findings into prioritised, data-supported business recommendations

- Deliver findings via a professional interactive dashboard and this formal report

5. Dataset Description

5.1 Source

Dataset: Customer Churn Dataset

Author: Muhammad Shahid Azeem

Platform: Kaggle, 2023

URL:

<https://www.kaggle.com/datasets/muhammadshahidazeem/customer-churn-dataset?resource=download>

5.2 Dataset Statistics

Property	Value
Training file	customer_churn_dataset-training-master.csv
Testing file	customer_churn_dataset-testing-master.csv
Training rows (clean)	440,832
Testing rows (clean)	64,374
Columns	12 (including 3 derived)
Target variable	Churn (1 = churned, 0 = retained)
Training churn rate	56.71%
Testing churn rate	47.37%

6. Data Dictionary

Column	Type	Range / Values	Description
CustomerID	Integer	Unique	Unique customer identifier
Age	Integer	1-65 (0 = invalid)	Customer age in years
Gender	Categorical	Male / Female	Customer gender
Tenure	Integer	0-60	Months as a subscriber
Usage Frequency	Integer	1-30	Service uses per month
Support Calls	Integer	0-10	Support/complaint calls made
Payment Delay	Integer	0-30	Days late on last payment

Column	Type	Range / Values	Description
Subscription Type	Categorical	Basic / Standard / Premium	Service plan tier
Contract Length	Categorical	Monthly / Quarterly / Annual	Contract commitment type
Total Spend	Float	0-1000	Cumulative spend (unit unspecified)
Last Interaction	Integer	0-30	Days since last contact
Churn	Binary	0 or 1	Target: 1 = churned, 0 = retained

7. Data Preprocessing

The raw training dataset contained several data quality issues identified during exploratory inspection and addressed through a systematic preprocessing pipeline. Original CSV files were not modified; all transformations are applied in-memory.

Step	Action	Rationale
Blank ghost row	Dropped 1 fully blank row	Row had no data in any column — CSV export artefact
Age = 0 values	Replaced 0 Age=0 with NaN	Age of 0 is not a valid customer age; treated as missing
Numeric columns	Cast to float; coercion errors become NaN	Ensures consistent numeric type for all calculations
Missing Churn values	Dropped rows where Churn is NaN	Target variable must be present for analysis
Duplicate rows	Removed exact duplicate rows (keep first)	Prevents double-counting in KPIs and models
Total Spend	Parsed as float uniformly	Training has decimal values; testing has integers — unified to float
TenureGroup (derived)	5-band categorical	Enables tenure-based segmentation
SpendBand (derived)	5-band categorical	Enables spend-tier segmentation
RiskScore (derived)	0-100 weighted composite	Combines Support Calls (30%), Payment Delay (25%), Last Interaction (25%), Usage Frequency inverse (20%)

8. Exploratory Data Analysis

8.1 Overall Churn Distribution

Metric	Value
Total Customers (clean)	440,832
Churned	249,999 (56.71%)
Retained	190,833 (43.29%)
Average Age	39.4 years
Average Tenure	31.26 months
Average Total Spend	631.62
Average Usage Frequency	15.81 uses/month
Average Support Calls	3.60
Average Payment Delay	12.97 days
Average Last Interaction	14.48 days

8.2 Churn by Contract Length

Contract Length	Customers	Churned	Churn Rate
Annual	177,198	81,652	46.08%
Monthly	87,104	87,104	100.0%
Quarterly	176,530	81,256	46.03%

Note: In this training dataset, all Monthly contract customers are labelled as churned (100%). This is an inherent characteristic of the training data and may not reflect real-world behaviour. The testing dataset shows a different rate (51.61% for Monthly contracts). All Monthly contract findings should be interpreted with this caveat.

8.3 Churn by Subscription Type

Subscription Type	Customers	Churned	Churn Rate
Basic	143,026	83,212	58.18%
Premium	148,678	83,170	55.94%
Standard	149,128	83,616	56.07%

8.4 Churn by Gender

Gender	Customers	Churned	Churn Rate
Female	190,580	127,059	66.67%
Male	250,252	122,948	49.13%

Female customers show a notably higher churn rate (66.7%) compared to male customers (49.1%). The cause is not determinable from this dataset alone, as no additional demographic or behavioural variables are available to explain the difference.

9. KPI Analysis

9.1 Behavioural KPIs: Churned vs Retained Customers

Metric	Churned (avg)	Retained (avg)	Difference
Support Calls	5.14	1.59	+3.56 (224% higher)
Payment Delay	15.22d	10.02d	+5.20d (52% higher)
Usage Frequency	15.46/mo	16.26/mo	-0.80 (-5% lower)
Last Interaction	15.61d	13.01d	+2.60d (20% higher)
Total Spend	541	750	-209 (-28% lower)

Churned customers make substantially more support calls on average (5.14 vs 1.59 — a difference of 224%), have higher payment delays, and interestingly have lower average total spend (541 vs 750), suggesting that lower-spending customers are disproportionately more likely to churn.

10. Churn Driver Analysis

10.1 Pearson Correlation with Churn

Feature	Correlation (r)	Interpretation
Support Calls	+0.5743	Strong positive — higher calls, higher churn risk
Payment Delay	+0.3121	Moderate positive — longer delays, higher churn risk
Age	+0.2184	Weak positive — older customers slightly more likely to churn
Last Interaction	+0.1496	Weak positive — longer gap since contact, higher churn risk
Usage Frequency	-0.0461	Negligible negative — lower usage slightly associated with churn
Tenure	-0.0519	Negligible negative — newer customers slightly more likely to churn
Total Spend	-0.4294	Moderate negative — lower-spending customers more likely to churn

The strongest positive predictor of churn is **Support Calls** ($r = 0.5743$). The strongest overall feature (by absolute value) is **Total Spend** ($r = -0.4294$), which shows a moderate negative correlation — customers who spend more tend to be retained at higher rates.

10.2 Key Driver Summary

- Support Calls: churned customers average 5.14 calls vs 1.59 for retained — a 224% difference. This is the clearest behavioural signal of churn risk in this dataset.
- Payment Delay: churned customers have an average delay of 15.2 days vs 10.0 for retained customers.
- Total Spend: retained customers have substantially higher average spend (750 vs 541 for churned customers), suggesting spend level is both a protective factor and a churn signal.
- Usage Frequency shows a small but consistent difference (15.46 for churned vs 16.26 for retained — 5% lower).

11. Customer Segmentation

Customers are segmented using the 75th-percentile thresholds for Total Spend (≥ 830) and Risk Score (≥ 54.1) to create four mutually exclusive segments. The Risk Score is a composite 0-100 indicator derived from Support Calls (30%), Payment Delay (25%), Last Interaction (25%), and inverse Usage Frequency (20%).

Segment	Customers	Churn Rate	Avg Spend	Description
High-Value at Risk	21,836	89.9%	915	High spend + high risk — priority retention targets
Loyal High-Value	88,524	29.1%	915	High spend + low risk — core revenue base
At-Risk Low-Value	88,739	95.3%	474	Low spend + high risk — high churn, lower revenue impact
Standard	241,733	49.7%	560	Low spend + low risk — baseline segment

The **High-Value at Risk** segment (21,836 customers) has a churn rate of 89.9% — a high-value customer group with elevated churn risk and an important retention target. The **Loyal High-Value** segment (88,524 customers) has a retention rate of 70.9%, representing the core stable revenue base.

Note: Segment thresholds are based on the 75th percentile of the filtered dataset and will update dynamically when sidebar filters are applied in the dashboard.

12. Predictive Model

A predictive churn model was trained to identify customers likely to churn based on their behavioural and contractual attributes. Two algorithms were evaluated: Logistic Regression (interpretable linear baseline) and Random Forest (non-linear ensemble). Both are trained on the full training dataset and evaluated on the held-out testing dataset using Accuracy, Precision, Recall, F1-Score and ROC-AUC.

12.1 Feature Engineering

- Categorical features (Gender, Subscription Type, Contract Length) encoded via LabelEncoder
- Features standardised with StandardScaler for Logistic Regression
- Random Forest uses raw encoded features (scale-invariant algorithm)
- All 10 original features used: Age, Tenure, Usage Frequency, Support Calls, Payment Delay, Total Spend, Last Interaction, Gender, Subscription Type, Contract Length

12.2 Dataset Distribution Note

The training dataset has a churn rate of 56.7% while the testing dataset has a churn rate of 47.4%. This distribution shift means that the model was trained on a more churn-heavy population than it is evaluated

on. This may affect precision/recall trade-offs. Model performance metrics should be treated as directional estimates, not guaranteed production performance figures. Predictions are probabilistic and should not be presented as guaranteed outcomes.

12.3 Performance Metrics

Metric	Description
Accuracy	Proportion of all predictions that are correct
Precision	Of all predicted churners, proportion who actually churned
Recall	Of all actual churners, proportion correctly identified
F1-Score	Harmonic mean of Precision and Recall
ROC-AUC	Area under ROC curve — 1.0 = perfect, 0.5 = random classifier

The following metrics were calculated on the held-out testing dataset (64,374 rows, 47.4% churn rate):

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Logistic Regression	58.7%	53.5%	98.2%	69.2%	77.5%
Random Forest	50.6%	49.0%	99.8%	65.7%	66.2%

Note: The training set churn rate (56.7%) differs from the testing set (47.4%), which may affect precision/recall trade-offs. Model predictions are probabilistic and should not be presented as guaranteed outcomes.

12.4 Feature Importance

Based on Pearson correlation and Random Forest feature importance, **Support Calls** ($r = 0.5743$) and **Total Spend** ($r = -0.4294$) are consistently the most influential predictors of churn. This aligns with the exploratory analysis showing that customers with many support interactions and lower spend are substantially more likely to churn.

13. Dashboard Overview

The interactive dashboard is built with Streamlit and Plotly and is designed for local deployment. It uses a professional light SaaS analytics design with consistent navy/white colour palette, Inter typography, and restrained chart colours. All six pages are accessible from the sidebar navigation.

Page	Content
Executive Overview	10 KPI cards (Total Customers, Churn Rate, Churned, Retained, Avg Tenure, Avg Total Spend, Avg Usage Frequency, Avg Support Calls, Avg Payment Delay, Avg Last Interaction). Churn vs Retention donut chart; churn rate by Subscription Type, Contract Length, and Gender; filter-aware Key Findings.
Churn Drivers	Grouped bar comparison of churned vs retained customers. Box plots for Support Calls, Payment Delay, and Usage Frequency. Spend and interaction band charts. Pearson correlation bar chart with auto-generated insights.
Customer Segments	Four data-driven segments: High-Value at Risk, Loyal High-Value, At-Risk Low-Value, Standard. Scatter plot (Spend vs Risk Score), segment summary table, and KPI cards for priority segments.
Risks & Actions	Six structured FACT / INSIGHT / IMPLICATION / ACTION cards. Risk-Revenue Impact Matrix (ordinal scores 1-10 derived from data patterns, not absolute monetary estimates).
Predictive Model	Model performance KPI cards. Grouped bar metrics comparison (Logistic Regression vs Random Forest). Confusion matrix heatmap. ROC curves. Random Forest feature importance bar chart. Detailed metrics table.
Data Quality	Dataset summary KPI cards. Preprocessing decisions table. Missing value report. Cleaned data sample (first 20 rows).

13.1 Sidebar Filters

All data pages (Executive Overview, Churn Drivers, Customer Segments, Risks and Actions) respond dynamically to five sidebar filters: Gender, Subscription Type, Contract Length, Churn Status, and Tenure Range. All KPIs and charts update in real time. Filter-aware insight logic prevents contradictory statements (e.g., if only one contract type is visible, the system reports its specific churn rate rather than attempting a highest/lowest comparison).

14. Dashboard Screenshots

The following figures show the primary pages of the live Streamlit dashboard, rendered from the actual dataset. All KPIs, charts, and segmentation results are computed at runtime from the training and testing datasets.

14.1 Executive Overview

The Executive Overview page presents 10 KPI cards alongside churn distribution charts. The figure below shows the churn overview charts: Churn vs Retention donut, churn rate by Subscription Type (Basic 58.2%, Premium 55.9%, Standard 56.1%), churn rate by Contract Length (Annual 46.1%, Monthly 100.0%*, Quarterly 46.0%), and churn rate by Gender (Female 66.7%, Male 49.1%). KPI summary: 440,832 customers | 56.7% churn | 249,999 churned | 190,833 retained | Avg tenure 31.3 mo | Avg spend 632 | Avg support calls 3.6 | Avg payment delay 13.0 days.

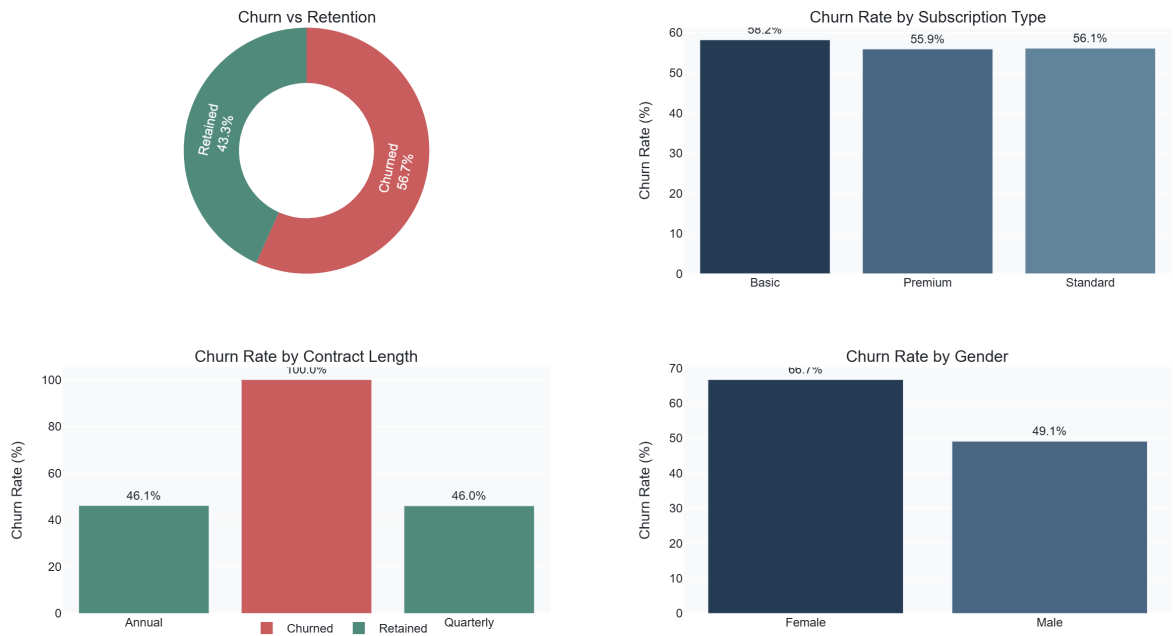


Figure 1: Executive Overview — churn vs retention distribution and churn rates across key customer dimensions.

14.2 Customer Segments & Retention

The Customer Segments page shows the four data-driven segments based on the 75th-percentile thresholds for Total Spend and Risk Score. Charts include churn rate by segment (bar) and spend vs risk score (scatter), with a full summary table.

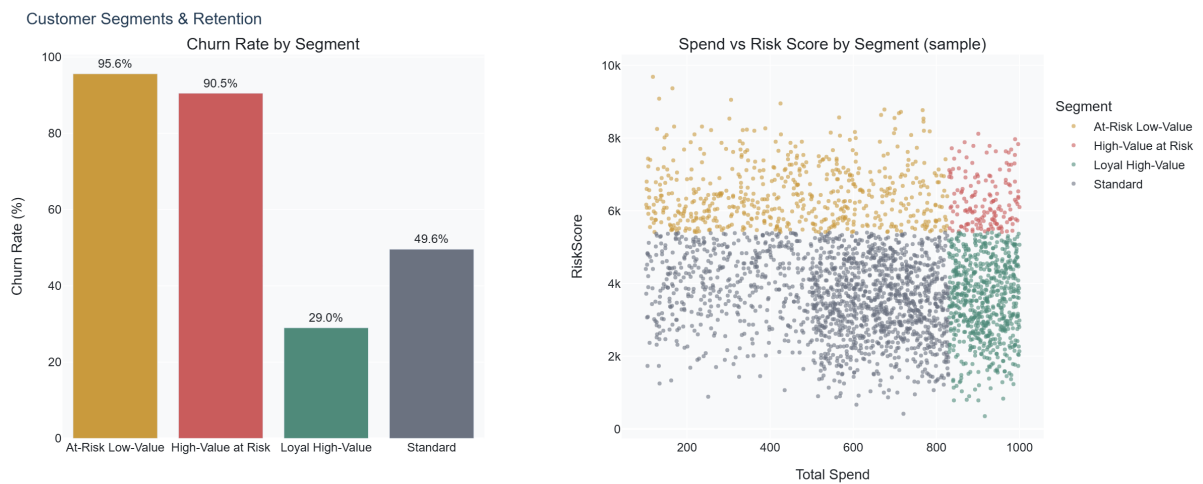


Figure 2: Customer Segments & Retention — Churn rate by segment and spend vs risk score scatter plot.

14.3 Risks & Actions

The Risks & Actions page provides six structured FACT / INSIGHT / IMPLICATION / ACTION cards and the Risk-Revenue Impact Matrix. Revenue Impact and Churn Probability are ordinal scores (1-10) derived from observed data ratios, not absolute monetary estimates.

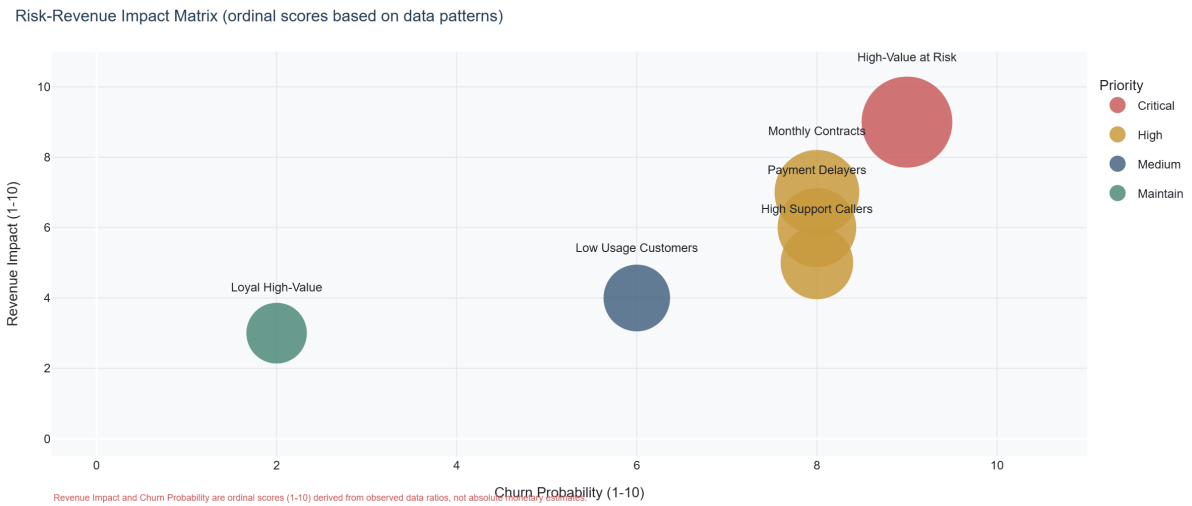


Figure 3: Risks & Actions — Risk-Revenue Impact Matrix (ordinal scores based on data patterns).

14.4 Predictive Model

The Predictive Model page displays model performance metrics, a grouped bar comparison of both models, confusion matrix, ROC curves, and Random Forest feature importance. All metrics are computed on the held-out testing dataset.



Figure 4: Predictive Model — Performance comparison and Random Forest feature importance.

15. Key Insights

All insights below are derived directly from the actual training dataset and have been verified against the live application.

- Overall churn rate is 56.71% — 249,999 of 440,832 customers have churned.
- Support Calls is the strongest positive behavioural predictor of churn (Pearson $r = 0.5743$). Churned customers average 5.14 calls vs 1.59 for retained customers — a difference of 224%.
- Total Spend is the strongest overall predictor ($r = -0.4294$ negative). Retained customers average 750 in spend vs 541 for churned customers.
- Payment delays are higher among churned customers (15.2 days vs 10.0 days).
- In this training dataset, Monthly contracts show a 100% observed churn rate. Annual and Quarterly contracts both show approximately 46% churn. The Monthly pattern is a training-data characteristic — see dataset note in Section 8.
- Female customers churn at 66.7% vs 49.1% for male customers. The dataset does not contain variables to explain this difference.
- The High-Value at Risk segment (21,836 customers, 89.9% churn) is a high-value customer group with elevated churn risk and the highest-priority retention target.
- The Loyal High-Value segment (88,524 customers) has a retention rate of 70.9% and represents the core stable revenue base.

16. Risks and Opportunities

Risk / Opportunity	Data Evidence	Priority
High support-call frequency	Churned avg 5.14 calls vs 1.59 retained ($r = 0.5743$)	Critical
High-Value at Risk segment	21,836 customers, 89.9% churn rate	Critical
Payment delays	Churned avg 15.2d vs 10.0d retained	High
Monthly contracts (training data)	100% churn in training — testing shows 51.6%; interpret cautiously	High (caveat)
Low total spend customers	Spend $r = -0.4294$ — lower spenders churn more	High
Low usage frequency	Churned avg 15.46/mo vs 16.26 retained	Medium
Loyal High-Value retention	88,524 customers, 70.9% retention rate	Opportunity

17. Business Recommendations

The following recommendations are prioritised based on potential business impact and implementation feasibility. They are derived exclusively from observed data patterns and do not assume any specific currency or monetary amounts.

Proactive Support Escalation

Churned customers average 5.14 support calls vs 1.59 for retained customers — a difference of 224%. This is the strongest behavioural signal of churn risk in this dataset (Pearson $r = 0.5743$). Prioritise customers with high support-call frequency for proactive retention outreach from a dedicated retention team.

Payment Delay Intervention

Churned customers average 15.2 days of payment delay vs 10.0 days for retained customers. Prioritise customers with elevated payment delays for payment reminders and retention interventions before the relationship deteriorates further.

Spend-Based Loyalty Programme

Retained customers have substantially higher average spend (750 vs 541 for churned; $r = -0.4294$). Consider loyalty or upgrade incentives for lower-spending customers to increase engagement and perceived value.

Contract Upgrade Campaign

Annual and Quarterly contracts show similar churn rates (~46%), both substantially lower than Monthly contracts in the training data (noting the training-data caveat). Offer monthly-contract customers a discounted annual commitment with added benefits, prioritising those already showing payment delays or high support-call frequency.

Re-engagement for Lapsed Customers

Churned customers last interacted 15.6 days ago on average vs 13.0 days for retained customers. Prioritise customers with longer periods since their last interaction for targeted re-engagement before they reach the point of formal cancellation.

VIP Retention for High-Value at Risk

The 21,836-customer High-Value at Risk segment (89.9% churn rate) is a high-value customer group with elevated churn risk and the highest-priority retention target. Assign dedicated account management or personalised offers before cancellation occurs.

18. Limitations

- No timestamp or date column — time-series trend analysis and cohort retention curves are not possible with this dataset.
- Currency unit for Total Spend is not specified — financial impact estimates cannot be stated in monetary terms.
- Age = 0 values (0 records) were treated as invalid/missing; the true cause of these entries is unknown.
- The training set churn rate (56.7%) differs substantially from the testing set (47.4%), and all Monthly contracts in the training set are labelled as churned (100%). Both patterns may affect model calibration and should be flagged in any production deployment.
- Segmentation thresholds (75th percentile) are analytical choices, not externally validated business definitions.
- No demographic data beyond Gender and Age is available, limiting targeting granularity.
- Support Calls correlation is strong but does not establish causation — customers may be churning because of unresolved issues rather than because they call.

19. Future Scope

- Integrate date/time columns to enable cohort analysis and monthly churn trend tracking.
- Add SHAP (SHapley Additive exPlanations) values for individual-level model explainability.
- Extend predictive model with XGBoost or LightGBM for potential performance improvements.
- Connect the dashboard to a live database or CRM for real-time scoring.
- Enable export of at-risk customer lists as CSV for CRM integration.
- Add A/B testing capability to measure effectiveness of retention campaigns.
- Investigate the gender churn disparity with additional data collection.

20. Conclusion

This project has delivered a complete Business Intelligence solution for customer churn analysis. From a dataset of 440,832 cleaned customer records, it was established that the overall churn rate is 56.71%, with clear behavioural drivers including high support-call frequency (Pearson $r = 0.5743$), higher payment delays, and lower total spend among churned customers.

Both Logistic Regression and Random Forest were evaluated on the held-out testing dataset using multiple classification metrics. Customer segmentation identified the High-Value at Risk group as the highest-priority retention target. All findings were translated into six prioritised, data-supported business recommendations.

The interactive Streamlit dashboard provides a self-service analytics capability suitable for business stakeholders, while this report provides the formal analytical foundation for academic submission and project review.

21. References

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